

BASAL ANGIOSPERMS

Of the more than 60 orders of flowering plants, or angiosperms, three, known as basal angiosperms, evolved early on and still exist today.

Modern DNA sequencing is a powerful way of identifying groups of species that are closely related to one another. However, the interrelationship between these groups is sometimes less clear, with different DNA markers suggesting different affinities. In flowering plants, there seems to be a pattern of some flourishing groups that have diversified hugely, leaving behind a few, remnant groups that have survived for millions of years but diversified little.

"Basal angiosperm" is the term used for three of these remnant orders. They are primitive species, apparently with little in common, growing

in widely scattered parts of the world. They include trees, shrubs, and climbers, as well as water plants. The Amborellales order survives today as a single species of shrub on one South Pacific island. The Nymphaeales are aquatic plants with primitive but often showy flowers. They include over 70 species of water lilies, found all over the globe. The Austrobaileales order contains nearly 100 species of woody plants, mainly in the tropics.

Two other orders feature on this page as "orphans", whose place in the phylogeny (evolutionary history) of angiosperm families is unclear. The most recent review suggests that the Chloranthales might be related to magnoliids, and the Ceratophyllales to eudicots, but for now that remains uncertain and unproven.

DIVISION	ANGIOSPERMAE
GROUPS	BASAL ANGIOSPERMS
ORDERS	3
FAMILIES	7
SPECIES	190

DEBATE MYSTERIOUS ORIGINS

Angiosperms may have evolved from seed ferns, a group that died out over 50 million years ago, or more probably from gnetophytes (see p.117). DNA and fossil evidence suggest that the Amborellales were the first angiosperms to appear, around 140 million years ago. The Nymphaeales and Austrobaileales were the next two groups to evolve.

CHLORANTHALES

The Chloranthaceae, the only family in the order Chloranthales, contains four genera and around 70 living species, but its fossil record dates back over 100 million years. Its members are mainly tropical shrubs and trees with small, stalkless flowers.

SARCANDRA GLABRA
F: Chloranthaceae

With a range of medicinal uses, this evergreen shrub inhabits damp ground, especially wooded banks of streams, in S.E. Asia, China, and Japan.



CERATOPHYLLALES

A single family, the Ceratophyllaceae, makes up this order. Called hornworts, these plants are completely unrelated to the non-flowering hornworts (see p.111). The four species are free-floating, rootless aquatics with whorls of finely-divided leaves, tiny male and female flowers, and spiny fruits.

RIGID HORNWORT
Ceratophyllum demersum
F: Ceratophyllaceae

This is a submerged species with no roots, inhabiting ponds and ditches in non-arctic Europe. It has tiny flowers and whorls of leaves.



AMBORELLALES

This order of primitive evergreen shrub contains a single family, which has a single genus with only one species, *Amborella trichopoda*. It bears small flowers, males and females being produced on separate plants, and red berries, each containing a single seed.

AMBORELLA TRICHOPODA
F: Amborellaceae

Found only on the island of New Caledonia in the South Pacific, this sprawling, white-flowered shrub is quite common in mountain forests.

AUSTROBAILEYALES

The order Austrobaileales is made up of only three families. Members of this order are trees, shrubs, and climbers. The flowers of most species are single and have many petals. Perhaps the best known species is the spice, star anise.



STAR ANISE
Illicium verum
F: Schisandraceae

Widely used for flavouring, star anise fruits are woody and star-shaped. This woodland species is native to China and Vietnam.

AUSTROBAILEYA SCANDENS
F: Austrobaileyaceae

The flowers of this rare primitive climber, found only in rainforests in Queensland, Australia, smell of rotting fish to attract pollinating flies.